

Walmart Sales Analysis

Business Insights Report

1. Executive Summary

This project analyzes Walmart's sales transaction data to understand **customer behavior, product performance, demand patterns, and branch-level profitability**.

Using SQL-based analysis on cleaned and feature-engineered data, the study identifies **key revenue drivers, peak demand periods, payment preferences, and performance risks**. The insights support data-driven decisions related to **staffing optimization, category strategy, payment infrastructure, and branch-level performance monitoring**.

2. Dataset Overview

- 9,969 sales transactions
- 100 branches across 98 cities
- 6 core product categories
- Invoice-level data with revenue, quantity, ratings, profit margin, and time attributes

The dataset provides sufficient scale and granularity for reliable operational and strategic analysis.

3. Key Business Insights

Customer Behavior

- Customers predominantly use **digital payment methods** (credit cards and e-wallets).
- Payment preferences vary across branches, indicating localized customer behavior.
- Most revenue is driven by **medium-value transactions**, not high spenders.

Insight: Revenue growth depends more on improving basket size and checkout efficiency than targeting a small high-spending segment.

Product & Profitability

- Revenue and profit are **highly concentrated in two product categories**, while others contribute marginally.
- High-revenue categories are not always the highest-margin ones.
- Customer satisfaction varies by **branch and city**, with different top-performing categories at each location.

Insight: Category and pricing strategies should balance revenue and profitability and be adapted at the branch level.

Time-Based Demand Patterns

- Sales peak during **afternoon and evening hours**, with mornings contributing the least.
- Certain weekdays consistently show higher demand.
- Monthly revenue fluctuates, indicating **seasonality rather than steady growth**.

Insight: Uniform staffing and promotions are inefficient; demand-driven scheduling is required.

Branch & Geographic Performance

- Each branch has a different busiest day, reflecting localized demand patterns.
- Some branches experienced **significant year-over-year revenue decline (50–63%)**.
- Profit margins vary notably by city, with some locations consistently outperforming others.

Insight: Early identification of underperforming branches and high-margin cities enables targeted operational and investment decisions.

4. Strategic Recommendations

1. Align staffing and inventory with **afternoon and evening peak demand**.
 2. Focus promotions and shelf space on **high-revenue and high-margin categories**.
 3. Encourage digital payments to improve checkout speed and reduce operational overhead.
 4. Investigate underperforming branches to identify and address root causes early.
 5. Use city-level profitability insights to guide pricing and expansion strategies.
 6. Increase average basket value through **bundling and upselling** targeted at medium-value customers.
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5. Limitations & Next Steps

Limitations

- No customer-level identifiers (limits retention analysis).
- No inventory or cost data (limits margin optimization).

Next Steps

- Customer cohort and repeat-purchase analysis
 - Promotion effectiveness and pricing sensitivity studies
 - Integration of inventory and cost data for deeper profitability analysis
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6. Conclusion

This analysis demonstrates how transactional sales data can be transformed into **actionable business insights**. By combining SQL analysis with business interpretation, the project highlights opportunities for **operational optimization, revenue growth, and risk mitigation**, making it relevant for real-world business decision-making in large retail organizations.